SYSTEM & AI ESSENTIALS • COMPREHENSIVE COURSE MANUAL

Systems, CLI, Git & Contest Master Guide

Complete In-Depth Conceptual Reference for Lectures 1 to 10, Exhaustive Command Cheat Sheets, and Rigorous Solutions for Contest 1 (50 Questions) & Contest 2 (25 Questions)

Core Systems Architecture

IPOS cycle, embedded vs servers vs supercomputers, CPU vs GPU parallelism, memory & storage hierarchies (RAM volatile vs SSD/HDD), device drivers, and OS resource management.

Terminal, Shell & Filesystem

Command execution, environment variables ($PATH resolution), Unix "Everything is a file" philosophy, pseudo-terminals (/dev/pts), permissions (chmod octal math & +x metadata).

Process Lifecycle & Execution

Program vs process, PID, isolated virtual address spaces, Chrome multi-process model (1 tab = 1 child process), CPU timeslicing, signals (kill -15 vs -9), and OOM handling.

Git & GitHub Distributed VCS

Working directory, Staging area, Local Repository (.git), commits, branching, merging, merge conflicts, remote origin, push, pull, fetch, clone, PRs, and GitHub collaboration.

Curriculum: System & AI Essentials (C01 - C10)

Includes: 75 Real Contest Questions & Detailed Explanations

Format: High-Yield Reference Edition

Module 1: Foundations of Computer Systems (Lecture 1)

1.1 The Fundamental Nature of a Computer

At its theoretical and practical core, a **computer** is an electronic machine that processes data according to a stored set of instructions. It operates along the classic **IPOS Model**:

- **Input:** Capturing raw data from the external physical environment (keystrokes, sensor readings, mouse clicks, packets).
- **Process:** Manipulating, computing, comparing, and transforming that data according to program logic executed by a central processing unit (CPU).
- **Output:** Delivering processed results into human-perceivable or machine-actuable form (screen pixels, radio transmissions, sound, printed sheets).
- **Storage:** Retaining instructions and state across immediate or extended periods (volatile RAM and non-volatile SSD/HDD).

> **The Defining Property: Reprogrammability vs Single-Purpose Tool**
> An orange juicer crushes oranges: it takes input, applies physical force, and gives juice. Yet it is *not* a computer. Why? Its behavior is permanently fixed into physical metal and gears.
> A digital scale with a microcontroller that weighs crates, receives a firmware update file over USB, and now calculates prices in Rupees and logs past weighings **is a computer**. Its behavior is dictated by **software**, and its hardware is general enough to carry out new, distinct sets of instructions without changing physical parts!

1.2 Taxonomy of Modern Computing Machines

Class	Primary Purpose	Key Characteristics	Real-World Example
Embedded System	Dedicated, single-purpose automated control	Microcontroller, fixed firmware, minimal/no OS, low power, extreme reliability, no user desktop	Water-level sensor box bolted to bridge; automobile ABS brakes; smart thermostat
Personal Computer (PC)	Interactive general-purpose computing for an individual	Desktop/laptop, rich graphical OS (macOS, Windows, Linux), mouse/keyboard/screen, runs diverse user software	District office desktop; student laptop; office workstation
Server	Serving centralized data and compute over a network to many clients	Rack-mounted, headless (no monitor/keyboard attached), optimized for multi-tenant throughput, 24/7 uptime	Rack-room machines fielding public flood-alert requests and telemetry database
Supercomputer	Massive parallel floating-point	Thousands of interconnected nodes, Petabytes of RAM, liquid cooled, specialized scheduling (SLURM)	Atmospheric simulation computing subcontinent weather grids; molecular

Class	Primary Purpose	Key Characteristics	Real-World Example
	mathematical simulations		dynamics; climate modeling

1.3 Failure Analysis: The Hazard of Unused Capability

In the weather department trial, 10 bridges replaced sealed embedded sensor boxes with ordinary Windows laptops. The laptops missed 43 readings, whereas the 190 simple boxes missed only 2. Every laptop failure stemmed from features the sensor task never needed: background Windows updates forcing restarts, power-saving sleep modes suspending the radio, and disk bloat from background logging.

> **Core Architectural Law:** Capability a job never uses is not free. It introduces points of failure, resource contention, and maintenance complexity. A general-purpose machine deployed where an embedded controller belongs is inherently unreliable and inefficient.

Module 2: Hardware Architecture & The Software Spectrum (Lecture 2)

2.1 Compute Engines: CPU vs GPU

Modern computers rely on specialized processing hardware tailored for distinct computational workloads:

- **Central Processing Unit (CPU):** The latency-optimized "brain" of the computer. Possesses a few powerful, highly sophisticated cores (typically 4 to 32) engineered to execute complex, sequential branch-heavy code at extremely high clock speeds with deep instruction-level parallelism.
- **Graphics Processing Unit (GPU):** The throughput-optimized SIMD (Single Instruction, Multiple Data) engine. Possesses thousands of smaller, simpler cores designed to execute the exact same arithmetic instruction simultaneously across millions of data points (e.g. image pixels, matrix multiplication in neural networks).

> **Diagnostic Clinic Case Study: The 40-Second vs 26-Minute Scan**
> Medical scan reconstruction executes identical arithmetic filtering on millions of pixels. The GPU's thousands of lightweight cores perform these operations in parallel in **40 seconds**. When the GPU was disabled, the CPU had to perform the operations sequentially through its few cores, taking **26 minutes**—even though the CPU itself was fast and healthy!

2.2 Memory and Storage Hierarchy

Layer	Technology	Speed / Latency	Volatility	Function in System
CPU Registers & Cache	SRAM (L1, L2, L3)	Sub-nanosecond to ~10 ns	Volatile (Lost on power cut)	Immediate instruction and data operands inside CPU die
RAM (Main Memory)	DRAM chips	~50 – 100 ns	Volatile (Lost on power cut)	Holds actively executing program code and live variables
Solid-State Drive (SSD)	NAND Flash Memory	~50 – 100 µs (Microseconds)	Non-Volatile (Retained permanently)	Secondary long-term storage; electronic silicon, high random IOPS
Hard Disk Drive (HDD)	Magnetic platters + arm	~5 – 15 ms (Milliseconds)	Non-Volatile (Retained permanently)	Secondary storage; mechanical spinning platters (slower, fragile)

> **Critical Speed Order:** `RAM > SSD > HDD`
> RAM is thousands of times faster than SSD, and SSD is up to 100 times faster than HDD due to zero mechanical seek time.

2.3 Memory Volatility and Power Cuts: The Unsaved Work Principle

When a user types a document, the active keystrokes and document state are held in **RAM**. RAM requires continuous electrical charge to maintain its state. When power drops, all capacitor charges in RAM dissipate immediately.

- *Saving a file* explicitly copies the buffer from volatile RAM into non-volatile secondary storage (SSD/HDD).
- Buying 64 GB of RAM does **not** protect unsaved work from a power outage: RAM loses its contents regardless of capacity. Only persisting to secondary storage guarantees durability.

2.4 Software Classification

- **System Software:** Software that provides the underlying platform and manages hardware resources so application software can run.
  - **Operating System:** Windows, macOS, Linux, Android (manages CPU, RAM, storage, security).
  - **Device Drivers:** Software adapters allowing the OS to communicate with specific hardware (printers, graphics cards, network cards).
  - **Utilities:** Antivirus tools, disk formatters, package managers.
- **Application Software:** Software created to accomplish specific end-user tasks: Web Browsers (Chrome), Spreadsheets (Excel), Word Processors (Word), Media Players (Spotify).

Module 3 & 4: Commanding the Machine & The Interface Continuum

3.1 Terminal, Shell, and the macOS Transition

The **Terminal** is an application (a graphical window emulator) that receives keyboard input and displays text output. The **Shell** is the command-line interpreter running inside the terminal that parses user commands, resolves paths, and executes programs.

- On modern macOS (macOS Catalina 10.15 and later), the default shell is **zsh** (Z Shell). Prior to 2019, macOS defaulted to **bash** (Bourne Again Shell).

3.2 Filesystem Navigation and Path Resolution

- **Root Directory (`/`):** The absolute top of the Unix filesystem hierarchy. Running `cd ..` at `/` leaves you at `/` because the root has no parent directory.
- **Absolute Path:** Starts from root (e.g. `/bin/ls` or `/Users/student/docs/file.txt`).
- **Relative Path:** Starts from the current working directory (e.g. `./script.sh` or `../folder/test.py`).

3.3 The `$PATH` Environment Variable

When you type a command like `ls` , the shell does not inherently know where the binary lives. It scans the colon-separated list of directories stored in the `$PATH` environment variable (e.g. `/usr/local/bin:/usr/bin:/bin:/usr/sbin:/sbin`).

> **What happens if $PATH is empty?**
> If `PATH=""` , typing `ls` outputs `command not found` because the shell cannot search any directory. However, **the program still exists** on disk! You can run it directly by specifying its absolute path: `/bin/ls` .

- `which <command>` : Prints the full absolute path of the executable resolved via `$PATH` (e.g. `which ls` → `/bin/ls`).
- `pwd` : Prints the current working directory path.

3.4 Wildcards & Command Construction

- `*` : Wildcard matching **zero or more characters**. Example: `rm -r folder_*` will delete `folder_1` , `folder_20` , and `folder_100` .
- `?` : Wildcard matching **exactly one character**. `folder_?` only matches single-digit folders like `folder_1` .
- `|` (Pipe): Feeds the standard output of the left command directly into standard input of the right command (e.g. `ps aux | grep chrome`).

3.5 Interfaces: CLI vs GUI vs API

Interface	Primary User	Input Mechanism	Strengths	Limitations
CLI (Command Line)	Developers, Sysadmins	Text commands & arguments	Scriptable, automatable, remote access, repeatable	Steep learning curve, memorization required
GUI (Graphical User)	General End Users	Mouse, touch, visual icons	Intuitive, discoverable, low barrier to entry	Hard to automate, heavy resource usage
API (Application Programming)	Software programs	HTTP requests, function calls, JSON	Seamless inter-system integration, standard contracts	Requires programming to utilize

Module 5: Operating Systems & Process Management (Lecture 5)

5.1 Program vs Process: A Crucial Distinction

- **Program:** A passive collection of instructions and static data stored as a file on secondary storage (SSD/HDD). Installed once.
- **Process:** An active instance of a program currently loaded into RAM and being executed by the CPU. A process contains an address space, stack, heap, CPU registers, file descriptors, and a unique **PID** (Process ID).

> **One Program → Multiple Independent Processes:** Launching three separate Google Chrome windows runs three or more distinct processes from the same single executable installed on disk.

5.2 Modern Multi-Process Architecture (Google Chrome Model)

Modern web browsers enforce process isolation: **1 Browser Tab = 1 Child Process**.

- **Why not threads or one big process?** If one tab crashes due to faulty JavaScript or a memory leak, only that specific child process terminates. The parent browser and all other tabs stay alive!
- **Memory Security:** Web tabs execute untrusted code from arbitrary domains. Process isolation prevents one tab from inspecting or altering the memory of online banking in an adjacent tab.

5.3 Virtual Memory and Address Space Isolation

Why cannot Google Chrome read or overwrite data held inside VS Code's memory?

> **Virtual Address Space Isolation:** The Operating System provides every process with its own private virtual address space mapped through hardware page tables. An address inside Chrome has no physical or logical relationship to that same numeric address inside VS Code. Process boundaries are strictly hardware-enforced by the MMU (Memory Management Unit).

5.4 Process Signals: Graceful Exit vs Immediate Termination

Command	Signal	Behavior	Use Case
`kill -15 <PID>`	**SIGTERM** (Signal 15)	Politely requests termination. The process intercepts the signal, closes open file handles, flushes network sockets, writes state, and exits cleanly.	Standard orderly shutdown
`kill -9 <PID>`	**SIGKILL** (Signal 9)	Kernel immediately kills the process without notifying it. Cannot be caught, blocked, or ignored. Unsaved state is destroyed.	Frozen, uncooperative, or runaway zombie processes

5.5 Out-Of-Memory (OOM) Errors: CPU vs Memory Allocation

When an application under heavy traffic crashes repeatedly with an **Out-Of-Memory (OOM) exception**:

- Adding more CPU cores will **not** fix the crash. CPU cores provide execution throughput; they do not expand memory capacity.
- The OS memory manager allocates chunks of RAM upon request. When physical RAM and swap space are exhausted, the OS cannot fulfill the request and invokes the OOM killer or throws a fatal memory exception.
- **The correct solution:** Increase system RAM or refactor the application to consume less memory.

Module 6: The Unix Filesystem Architecture & Permissions (Lecture 6)

6.1 The Unix Philosophy: "Everything is a File"

In Unix-like operating systems (macOS, Linux), hardware devices, communication channels, and processes are exposed through the standard filesystem interface:

- `/dev/pts/*` (Pseudo-Terminals): Terminals appear as files. Running `echo "hi" > /dev/pts/2` writes the string "hi" directly onto the screen of terminal 2!
- `/dev/input/*`: Mouse and keyboard event streams represented as readable files.
- `/tmp`: Temporary workspace for transient files. **Automatically wiped clean on system reboot!**
- `/bin`: Essential system binary executables needed for single-user mode and booting.
- `/etc`: Host-specific system-wide configuration files.

6.2 File Permission Triplets and Octal Calculation

Running `ls -l` displays a 10-character string such as `-rw-r--r--`:

```
-  r w -  r - -  r - -
^  [Owner] [Group] [Others]
Type: (-) Regular file, (d) Directory
```

Permission	Symbol	Numeric (Octal) Value	Meaning on Files	Meaning on Directories
Read	`r`	**4**	View file content	List contents (run `ls`)
Write	`w`	**2**	Modify / truncate file	Create / delete files in dir
Execute	`x`	**1**	Run file as a program	Enter / `cd` into directory

Master Permission Matrix & Key Values

- **`chmod 644`** (`-rw-r--r--`)**:** Owner has Read + Write (4+2=6); Group has Read (4); Others have Read (4). Standard permission for text files, web pages, and public documents.
- **`chmod 600`** (`-rw-------`)**:** Owner has Read + Write (4+2=6); Group has None (0); Others have None (0). Strict privacy for SSH private keys and personal notes.
- **`chmod 754`** (`-rwxr-xr--`)**:** Owner has Read + Write + Execute (4+2+1=7); Group has Read + Execute (4+1=5); Others have Read (4).
- **`chmod 755`** (`-rwxr-xr-x`)**:** Owner has Read + Write + Execute (7); Group and Others have Read + Execute (5). Standard for executable scripts and directories.
- **`chmod 777`** (`-rwxrwxrwx`)**:** Full permissions for everyone. *Extremely dangerous anti-pattern.* Grants write and execute access to any arbitrary user on the machine.

6.3 Execution Metadata vs File Extensions

- On Windows, a file's executable status is tied to its extension (`.exe` , `.bat`).
- On Unix/Linux/macOS, execution is determined **strictly by the execute bit (`x`) stored in file metadata**.
- A file named `hello.txt` made executable via `chmod +x hello.txt` and renamed to `run_me.banana` will execute cleanly with `./run_me.banana` because the metadata bit persists across renames!
- **Why newly created files lack execute permission:** Security safeguard to prevent unvetted downloaded scripts from accidentally running.

Module 7 & 8: Software Execution & Low-Level Data Representation

7.1 Software Execution Pipeline: Compilers vs Interpreters

The CPU understands only binary machine code. Every programming language must eventually resolve to machine code:

- **Compiled Languages (C, C++, Rust, Go):** An Ahead-Of-Time (AOT) compiler reads source code, optimizes it, and emits a standalone binary machine code file tailored directly for the target CPU architecture. Ultra-fast execution, zero runtime interpreter overhead.
- **Interpreted Languages (Python, Ruby, JavaScript):** Source code is executed via an interpreter program (e.g. `/usr/bin/python3`). When you run `python3 hello.py`, two programs are involved: the Python interpreter binary and your script. The interpreter parses code and evaluates it dynamically. Portable, but slower than compiled code.
- **Just-In-Time (JIT) Compilation:** Hybrid approach (e.g. PyPy, Java HotSpot, V8) that compiles frequently executed bytecode loops into native machine code at runtime.

7.2 Low-Level Data Representation: Bits, Bytes, and Hex

- **Bit:** The fundamental binary unit (0 or 1), physically represented by electrical voltage levels (high vs low).
- **Byte:** 8 bits. Represents $2^8 = 256$ distinct values (0 to 255 unsigned, or -128 to +127 signed).
- **Hexadecimal (Base-16):** Uses digits `0-9` and letters `A-F` (where A=10, B=11, C=12, D=13, E=14, F=15).
  - 1 Hex digit = 4 binary bits (1 Nibble). E.g., binary `1111` = Hex `F`.
  - 1 Byte = Exactly 2 Hex digits. E.g., binary `11111111` = Hex `FF` (decimal 255).

7.3 Text Encodings: ASCII vs Unicode UTF-8

- **ASCII:** 7-bit standard (128 characters) covering standard English alphabet (A-Z, a-z, 0-9, punctuation, control codes). E.g., 'A' is 65 (binary `01000001`).
- **Unicode (UTF-8):** Variable-length encoding (1 to 4 bytes) that represents over 150,000 characters spanning all human languages, mathematical symbols, and emoji. Backward-compatible with ASCII in its first 128 code points.

7.4 Image Representation & File Signatures

- **RGB Image:** Each pixel is composed of 3 bytes: Red (0–255), Green (0–255), Blue (0–255).
- **RGBA Image:** Adds an Alpha transparency channel (4 bytes per pixel).
- **Uncompressed Image Size:** $ ext{Size (bytes)} = ext{Width} imes ext{Height} imes ext{Channels}$. A $1920 imes 1080$ RGB image uncompressed requires $ pprox 6.22 ext{ MB}$.

- **Magic Numbers (File Signatures):** Unique byte sequences at the beginning of files that identify their format (e.g., PNG files start with bytes `89 50 4E 47 0D 0A 1A 0A`).
- **Encoding vs Encryption:** Encoding converts data into standard formats for interoperability (public algorithm, easily reversed without a key). Encryption scrambles data using mathematical ciphers and secret keys to enforce confidentiality.

Module 9 & 10: Version Control Systems — Git & GitHub (Lectures 9 & 10)

9.1 The Git Architecture: The Three Trees

Git is a distributed version control system that runs entirely on your local machine inside a hidden `.git` repository directory:

```
[ Working Directory ]  --- git add --->  [ Staging Area (Index) ]  --- git commit --->  [ L
(Actual modified files)                   (Snapshot prepared for commit)                 (P
```

- **Working Directory:** The active filesystem folder where you edit files. Files can be *untracked* or *modified*.
- **Staging Area (Index):** The intermediate staging buffer where changes are selected and reviewed prior to recording.
- **Local Repository (Commit History):** The cryptographically verified Directed Acyclic Graph (DAG) of commit snapshots stored inside `.git`.

9.2 Git vs GitHub: Local Tool vs Remote Hub

Dimension	Git	GitHub
Nature	Command-line Version Control tool	Cloud hosting platform & collaboration hub
Location	Installed locally on your personal machine	Hosted on cloud servers (github.com)
Function	Tracks file changes, commits, branches locally	Coordinates team sharing, Pull Requests, code reviews, CI/CD
Internet	Functions 100% offline without network	Requires internet connectivity to push/pull/browse

9.3 The Core Collaboration Workflow

1. **Initialize & Configure:**

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
git init
```

2. **Stage & Commit Changes:**

```
git status               # Check working tree status
git add notes.txt        # Move file to Staging Area
git commit -m "feat: add initial notes"  # Commit to local repo
```

3. **Connect to Remote GitHub Repository:**

```
git remote add origin https://github.com/username/repo.git
git branch -M main
git push -u origin main   # Send commit history to GitHub
```

4. **Team Collaboration & Pull Requests:**

```
git clone https://github.com/org/repo.git   # Download full repo
git switch -c feature-branch                # Create and switch to new branch
git push origin feature-branch              # Push branch to remote
# Open a Pull Request (PR) on GitHub for code review, then Merge into main!
```

Master CLI & Git Command Reference

Essential Shell & Terminal Commands

Command	Description	Example Usage
`pwd`	Print Working Directory (prints full absolute path)	`pwd`
`ls`	List directory contents	`ls -la` (all files, hidden, long format)
`cd <dir>`	Change directory (`cd ..` moves up, `cd ~` home)	`cd ../project`
`mkdir <dir>`	Create a new directory	`mkdir -p parent/child`
`touch <file>`	Create empty file or update timestamp	`touch app.py`
`cp <src> <dst>`	Copy files (use `-r` for directories)	`cp -r src/ backup/`
`mv <src> <dst>`	Move or rename files and directories	`mv old.txt new.txt`
`rm <file>`	Remove file (use `-r` for directories)	`rm -r folder_*`
`cat <file>`	Concatenate and display file content	`cat notes.txt`
`echo <str>`	Print text to standard output	`echo "hello" > /dev/pts/1`
`grep <pattern>`	Search for text matching regex pattern	`grep -rn "TODO" ./src`
`which <cmd>`	Locate executable path resolved via `$PATH`	`which ls` (→ `/bin/ls`)
`ps aux`	Display snapshot of all running processes	`ps aux | grep python`
`top` / `htop`	Interactive real-time process viewer	`top`
`kill -15 <pid>`	Send graceful termination signal (SIGTERM)	`kill -15 1420`
`kill -9 <pid>`	Force immediate uncatchable process kill (SIGKILL)	`kill -9 1420`
`chmod <mode>`	Change file permissions	`chmod +x run.sh` , `chmod 644 doc.txt`
`chown <owner>`	Change file ownership (requires `sudo`)	`sudo chown root:admin config.cfg`

Complete Git & GitHub Command Reference

Command	Function & Purpose	Key Flags & Variations
`git init`	Initialize a brand new local Git repository	Creates hidden `.git/` directory in project root
`git config`	Set author credentials and preferences	`git config --global user.name "Name"`
`git status`	Inspect working tree & staged files status	`git status -s` (short output format)
`git add <file>`	Add file modifications to the Staging Area	`git add .` (stages all tracked/untracked changes)
`git commit -m <msg>`	Permanently record snapshot in repository history	`git commit -am "msg"` (stages tracked & commits)
`git log`	View chronological history of commits	`git log --oneline --graph --decorate`
`git diff`	View line-by-line differences	`git diff --staged` (compares staged vs last commit)
`git branch`	List, create, or delete branches	`git branch -M main`, `git branch -d old-feat`
`git switch` / `checkout`	Switch between branches or restore files	`git switch -c feature` (create and checkout branch)
`git merge <branch>`	Integrate another branch into the current branch	`git merge feature-login`
`git remote add`	Link local repository to a remote GitHub URL	`git remote add origin https://github.com/...`
`git push`	Upload local commits to remote GitHub repository	`git push -u origin main` (sets upstream tracking)
`git pull`	Fetch and integrate remote changes into local branch	`git pull origin main` (equivalent to fetch + merge)
`git clone <url>`	Download complete remote repository with full history	`git clone https://github.com/owner/repo.git`

Contest 1: 50 Real Questions & In-Depth Solutions

Detailed analysis of all 50 questions covering computer classification, hardware, memory hierarchy, operating systems, and process management.

Question 1 / 50

CORRECT ANSWER: A

Aman wants faster boot times and quicker app loading on his old laptop. Which upgrade would be most effective?

- **Replacing his HDD with an SSD**
- Buying a bigger monitor
- Getting a mechanical keyboard
- Adding external speakers

Explanation: Boot time and app loading depend heavily on how fast the storage device can read data. SSDs are many times faster than HDDs because they have no moving parts, so switching to an SSD gives a huge speed boost.

Question 2 / 50

CORRECT ANSWER: B

Which of the following is a primary responsibility of an operating system?

- Creating spreadsheets
- **Managing system resources**
- Designing websites
- Editing photographs

Explanation: An operating system acts as the core resource manager of the machine, coordinating CPU time, memory allocation, storage access, and I/O devices among running processes.

Question 3 / 50

CORRECT ANSWER: B

A state weather department is building a flood-warning system. Four kinds of machine are involved:
(1) Waterproof boxes bolted to bridges with small board, water-level sensor, radio.
(2) District office machines on desks where officers write reports.
(3) Rack-room machines with no person sitting in front of them that receive readings and serve citizen requests.
(4) National centre machine running atmospheric simulation dividing the sky into billions of cells.
Which classification of machines (1) to (4) matches the lecture categories?

- (1) personal computer, (2) embedded system, (3) supercomputer, (4) server
- **(1) embedded system, (2) personal computer, (3) server, (4) supercomputer**
- (1) embedded system, (2) server, (3) personal computer, (4) supercomputer
- (1) server, (2) personal computer, (3) embedded system, (4) supercomputer

Explanation: (1) Dedicated sensor board = Embedded system; (2) Interactive officer workstation = Personal computer; (3) Headless multi-user data provider = Server; (4) Massive parallel atmospheric simulation = Supercomputer.

Question 4 / 50

CORRECT ANSWER: C

Which feature of the rack-room machines - rather than their physical size or their speed - is what puts them in their category?

- They are kept in a locked, climate-controlled room.
- They have more RAM installed than the machines in the district office.
- **They store data and provide it as a service to many other machines at the same time, over a network.**
- They need no operating system of their own, because no person ever sits in front of them to run a program.

Explanation: A server is defined by its role: providing centralized data, compute, or network services concurrently to multiple remote clients over a network.

Question 5 / 50

CORRECT ANSWER: D

In a trial, 10 bridge boxes are replaced with ordinary laptops. Over one monsoon the laptops miss 43 readings due to Windows updates, power-saving sleep, and disk bloat; the 190 embedded boxes miss only 2. Which conclusion do these findings support?

- The laptops needed better weatherproofing.
- The laptops needed faster processors and more memory.
- The boxes are more reliable only because they were installed first.
- **Capability a job never uses is not free. It can become a cause for which the job can fail, which is why a computer meant for one task is inefficient here.**

Explanation: Unused features (desktop OS background services, auto-sleep, automatic updates) act as failure vectors. A dedicated embedded system has zero superfluous background tasks.

Question 6 / 50

CORRECT ANSWER: A

Halfway through the season the department decides the river boxes should measure every ten seconds instead of thirty, and report battery level. Assuming the boxes allow it, what does the definition of a computer predict?

- **The behaviour can be changed by loading a new set of instructions into the hardware already on the bridge.**
- Every box must be physically replaced, because its behaviour is fixed by the way it was built.
- Only the national centre's machine can be reprogrammed.
- Each box needs a GPU added before it can handle battery-level data.

Explanation: A computer's defining trait is reprogrammability: you update software instructions without altering physical hardware.

Question 7 / 50

CORRECT ANSWER: C

During a flood, the public website becomes unreachable for citizens. At the same time, officers in district office can read data on their screens and bridge boxes transmit normally. Which layer is the most likely point of failure?

- The river boxes - they are exposed to extreme weather.
- The national centre's simulation.
- **The rack-room machines' capacity to answer many simultaneous requests – failure appears where users rose sharply.**
- The officers' desk machines.

Explanation: Because internal office views and river sensors function normally, the bottleneck is exclusively at the public-facing server layer overwhelmed by concurrent citizen traffic.

Question 8 / 50

CORRECT ANSWER: B

Rahul needs to carry a 2 GB project file from home to college to submit it. Which device is the most convenient choice?

- The internal HDD of his desktop
- **A USB flash drive**
- His computer's monitor
- A RAM stick

Explanation: A USB flash drive is compact, non-volatile, portable storage. RAM loses data without power, monitors do not store data, and internal HDDs are non-portable.

Question 9 / 50

CORRECT ANSWER: C

A computer becomes almost unusable because many programs are running simultaneously and consuming available memory. Which OS function is directly involved?

- Document formatting
- Screen resolution
- **Memory management**
- File naming

Explanation: Memory management controls RAM allocation, paging/swapping to disk, and reclaiming memory from terminating processes.

Question 10 / 50

CORRECT ANSWER: D

Which software would normally be used to protect a computer by detecting and removing malicious programs?

- Spreadsheet software
- Word processor
- Presentation software
- **Antivirus utility**

Explanation: Antivirus software is a system maintenance utility specialized in detecting, quarantining, and eliminating malware.

Question 11 / 50

CORRECT ANSWER: D

Which software would be most appropriate for calculating monthly expenses and creating charts from the results?

- Antivirus utility
- Operating system
- Device driver
- **Spreadsheet application**

Explanation: Spreadsheets (like Microsoft Excel, Google Sheets) are specifically designed for grid calculations and data visualization.

Question 12 / 50

CORRECT ANSWER: C

A computer has enough memory and storage, but a newly connected printer does not work because the OS cannot communicate with it. What is missing?

- Web browser
- File manager
- **Device driver**
- Spreadsheet

Explanation: A device driver translates generic OS print requests into hardware-specific control codes the printer understands.

Question 13 / 50

CORRECT ANSWER: B

Why is an SSD generally faster than an HDD?

- It uses larger spinning disks
- **It has no moving parts**
- It uses magnetic tape
- It is physically bigger in size

Explanation: SSDs utilize electronic NAND flash memory with zero mechanical seek time, unlike HDDs which require moving magnetic platters and read/write heads.

Question 14 / 50

CORRECT ANSWER: B

Front-desk machine: modest CPU, 8 GB RAM, 1 TB HDD. Imaging machine: fast CPU, 32 GB RAM, 1 TB SSD, dedicated GPU. Front-desk machine takes 2 minutes to open patient-records; imaging machine takes seconds. Why?

- Front-desk machine has less RAM, where programs are kept permanently.
- **Front-desk machine reads from spinning platters and moving heads; imaging machine reads from flash memory with no moving parts.**
- Processor speed alone controls drive reading.
- The age of a drive alone determines speed.

Explanation: Application launch speed is I/O-bound by secondary storage read rates. SSD flash memory transfers binaries orders of magnitude faster than mechanical HDDs.

Question 15 / 50

CORRECT ANSWER: D

During a power cut, a receptionist lost the half-typed record she had not yet saved. Every saved record was intact when power returned. What does this demonstrate?

- The HDD is volatile, losing unsaved data.
- Unsaved work was held inside the CPU registers.
- 8 GB RAM was too small to hold the record.
- **Work being typed lives in RAM, which is volatile; saving copies it into secondary storage, which keeps data without power.**

Explanation: Active memory lives in volatile DRAM. Only flushing/writing to non-volatile secondary storage (HDD/SSD) preserves data across power failures.

Question 16 / 50

CORRECT ANSWER: A

The imaging machine reconstructs a scan in 40 seconds. When the graphics card is disabled, the same scan takes 26 minutes, even though the CPU was untouched. Why?

- **The reconstruction applies the same operation repeatedly, and a GPU's thousands of small cores handle repetitive parallel work far better than a CPU's few powerful cores.**
- The graphics card holds its own copy of the OS.
- A CPU cannot perform arithmetic directly.
- Disabling the graphics card disabled the SSD.

Explanation: Image reconstruction is massively parallel pixel math (SIMD). GPUs excel at parallel repetitive arithmetic across thousands of cores; CPUs must process them sequentially through a few cores.

Question 17 / 50

CORRECT ANSWER: C

The clinic can upgrade exactly one component on the front-desk machine to fix slow 2-minute application startup. Which upgrade should it choose?

- Add a graphics card
- Double RAM to 16 GB
- **Replace the hard disk drive with an SSD, because the complaint is about how long data takes to be read out of long-term storage.**
- Replace the processor

Explanation: Program startup is bounded by secondary storage read throughput. Replacing the mechanical HDD with an SSD eliminates the 2-minute loading bottleneck.

Question 18 / 50

CORRECT ANSWER: D

A vendor claims: "Buy 64 GB of RAM and your staff will never again lose unsaved work in a power cut." How should the clinic judge this claim?

- Accept it - larger RAM keeps data clear of overwritten parts.
- Accept it - above 32 GB RAM behaves like an SSD.
- Reject it - RAM cannot hold documents at all.
- **Reject it - RAM loses its contents when power is lost regardless of how much of it there is; only writing to secondary storage protects the work.**

Explanation: Volatility is an intrinsic physical characteristic of DRAM. Capacity does not alter volatility.

Question 19 / 50

CORRECT ANSWER: A

Which pair contains only application software?

- **Chrome and Excel**
- Windows and macOS
- Linux and Android
- BIOS and Linux

Explanation: Chrome (browser) and Excel (spreadsheet) are user applications. Windows, macOS, Linux, and Android are operating systems; BIOS is firmware.

Question 20 / 50

CORRECT ANSWER: C

A monitor is an example of which type of device?

- Input device
- Storage device
- **Output device**
- Processing device

Explanation: A monitor presents visual output generated by the computer's graphics hardware.

Question 21 / 50

CORRECT ANSWER: C

Machine A: mechanical juicer. Machine B: digital scale updated via USB file to calculate rupee prices and store past 20 weighings. Machine C: billing laptop. Meera says: "All three take input, do something, give output, so all three are computers." Which property is missing from her argument?

- The machine must be powered by electricity.
- The machine must produce output faster than humans.
- **The machine's behaviour must be changeable by loading a different set of instructions into it.**
- The machine must have a display.

Explanation: Reprogrammability—the ability to execute different instruction sets without altering the underlying physical machinery—is the hallmark of a computer.

Question 22 / 50 CORRECT ANSWER: A

Applying that reprogrammability property as the test, which classification of Kiran's three machines is correct?

- **Machines B and C are computers; Machine A is not.**
- Only Machine A is a computer.
- All three are computers.
- Only Machine C is a computer.

Explanation: Machine B (firmware reprogrammable digital scale) and Machine C (laptop) execute stored instructions that can change; Machine A is pure fixed mechanical hardware.

Question 23 / 50 CORRECT ANSWER: B

Machine B gained new behaviour - price display and past weighings memory - without a single physical part replaced. What does this demonstrate?

- Machine B's hardware is more powerful than A's motor.
- **What Machine B does is determined by its software, and its hardware is general enough to carry out more than one set of instructions.**
- Machine B has more secondary storage than A.
- Machine B is an application software rather than hardware.

Explanation: Demonstrates the separation of hardware and software: software directs behavior over general execution hardware.

Question 24 / 50 CORRECT ANSWER: D

Kiran asks for a new scale feature: a distinct beep whenever a crate is lighter than the crate weighed immediately before it. What does this feature require that the first one does not?

- An output device
- A faster processor
- A second sensor
- **A value kept from an earlier measurement, because the machine must compare the current input against something no longer on the pan.**

Explanation: Comparing the current item against a previous item requires memory storage to preserve past state across measurements.

Question 25 / 50

CORRECT ANSWER: C

A student says: "Machine C is a real computer and Machine B is only a gadget, because Machine C runs many programs while Machine B runs one." What is the strongest objection?

- Machine B has a display.
- Machine C is an embedded system.
- **How many programs a device is currently made to run is a design decision; the test is whether it can be given different instructions, and Machine B was shown to pass that test.**
- Machine B weighs fruit faster.

Explanation: The qualification of a computer rests on capability for instruction-driven execution, not whether its creator chose to limit it to one domain.

Question 26 / 50

CORRECT ANSWER: B

Riya's laptop slows down whenever she opens too many browser tabs, Spotify, and Word at once. Upgrading which component would help most?

- Buying a bigger monitor
- **Increasing the RAM**
- Getting a wireless mouse
- Adding more USB ports

Explanation: Multiple concurrent active applications demand RAM. When RAM is exhausted, the OS resorts to slow disk paging, degrading responsiveness.

Question 27 / 50

CORRECT ANSWER: C

A student wants to write assignments, calculate marks, and create presentations. Which type of software will be most useful?

- Device drivers
- Firmware
- **Application software**
- System software

Explanation: Office suites (word processors, spreadsheets, presentation creators) are application software.

Question 28 / 50 CORRECT ANSWER: A

Which option correctly matches the software with its category?

- **Android — Operating system**
- Linux — Application software
- Excel — System software
- Chrome — Device driver

Explanation: Android is a mobile operating system based on the Linux kernel.

Question 29 / 50 CORRECT ANSWER: B

Which of the following is a characteristic of system software?

- It is used only for internet browsing
- **It supports the operation of the computer**
- It is designed only for presentations
- It is created only for entertainment

Explanation: System software manages hardware resources and provides foundational execution environments for applications.

Question 30 / 50 CORRECT ANSWER: B

Speakers are considered a(n) _____ device.

- Input
- **Output**
- Storage
- Processing

Explanation: Speakers convert electrical audio waveforms from the computer into physical sound waves for human hearing.

Question 31 / 50 CORRECT ANSWER: B

What does SSD stand for?

- Simple Storage Device
- **Solid State Drive**
- System Storage Disk
- Secondary Save Drive

Explanation: Solid State Drive — called "solid state" because it relies entirely on solid semiconductor chips without moving parts.

Question 32 / 50 CORRECT ANSWER: A

A lab has identical hardware dual-booting Windows and Ubuntu. Why do the two sessions behave and look so differently despite identical hardware?

- **The operating system sits between user and hardware; changing it changes how the machine is presented and controlled, while hardware is unchanged.**
- Choosing an OS changes CPU clock speed.
- Ubuntu reconfigures the SSD into a different physical format.
- The OS divides RAM differently, which dictates screen design.

Explanation: The OS acts as the mediator and abstraction layer between user software and raw electronic hardware.

Question 33 / 50 CORRECT ANSWER: B

Three pieces of software: browser, word processor, package manager. Which grouping separates system software from application software correctly?

- System: word processor. App: package manager and browser.
- **System software: package manager. Application software: browser and word processor.**
- System: browser and package manager. App: word processor.
- All four are application software.

Explanation: A package manager manages system utilities and library dependencies (system software); browsers and word processors are user productivity tools (application software).

Question 34 / 50

CORRECT ANSWER: D

A browser needs to write a downloaded file onto the SSD. Using the three-layer stack, what actually happens?

- The browser writes to SSD directly.
- The SSD runs the browser.
- The compiler must translate the file first.
- **The browser asks the operating system for the service, and the operating system, which manages the hardware, carries out the write.**

Explanation: User applications cannot directly access disk hardware. They make system calls to the OS kernel, which validates permissions and writes through storage drivers.

Question 35 / 50

CORRECT ANSWER: C

A new printer prints immediately in Windows, but under Ubuntu prints nothing even though connected. What is the most reasonable explanation?

- The printer hardware is incompatible.
- Ubuntu has no applications installed.
- **The two systems need different device drivers, and the driver for this printer is present in one system's software collection but not in the other's.**
- Ubuntu lacks a GPU.

Explanation: Hardware devices require OS-specific device drivers. Windows had the driver pre-installed, whereas Ubuntu lacked it.

Question 36 / 50

CORRECT ANSWER: A

Bhaskar claims the Ubuntu tool to view running programs and Windows Task Manager are "the same thing with a different name". Which is the most defensible version of this claim?

- **Both are supplied by their operating system to report how the machine's resources – memory and processor time – are being used.**
- Both are applications written by the same company.
- Both are device drivers.
- Both are compilers.

Explanation: Both serve the identical system utility function: querying the OS kernel to inspect process states and hardware resource utilization.

Question 37 / 50

CORRECT ANSWER: B

A student says: "My computer has 512 GB of RAM to store all my movies and games permanently." What's wrong with this statement?

- Nothing — RAM stores files permanently
- **RAM is temporary storage**
- 512 GB is too little for storage
- RAM cannot hold any data at all

Explanation: RAM is volatile memory that loses all state upon power cutoff. Permanent files must be saved on secondary storage (SSD/HDD).

Question 38 / 50

CORRECT ANSWER: D

Which of these is NOT an input device?

- Keyboard
- Mouse
- Microphone
- **Speaker**

Explanation: Keyboards, mice, and microphones transmit data into the computer. Speakers output audio from the machine.

Question 39 / 50

CORRECT ANSWER: B

Which statement best describes computer software?

- Electrical signals inside a computer
- **Instructions used by a computer**
- Physical components of a computer
- Connections between hardware devices

Explanation: Software is the organized collection of programs, scripts, and instructions directing hardware execution.

Question 40 / 50

CORRECT ANSWER: B

Which of the following best describes a Hard Disk Drive (HDD)?

- A drive with no moving parts
- **A storage device with spinning disks**
- A temporary memory used by the CPU
- A device used only for display

Explanation: HDDs store bits magnetically across spinning mechanical platters accessed via an actuator arm.

Question 41 / 50

CORRECT ANSWER: C

What happens to the data stored in RAM when you turn off your computer?

- It stays saved forever
- It gets backed up to cloud
- **It is erased and lost**
- It moves to hard disk automatically

Explanation: RAM requires electrical charge to maintain state. Power interruption dissipates all stored bits.

Question 42 / 50

CORRECT ANSWER: B

Arrange these from fastest to slowest in terms of data access speed:

- HDD → SSD → RAM
- **RAM → SSD → HDD**
- HDD → RAM → SSD
- SSD → HDD → RAM

Explanation: RAM operates in nanoseconds, SSD in microseconds, and mechanical HDD in milliseconds.

Question 43 / 50

CORRECT ANSWER: C

A user saves documents inside folders and later searches for a particular file. Which operating system capability is being used?

- Memory allocation
- Device installation
- **File management**
- Process scheduling

Explanation: Filesystem management organizes directories, path resolutions, and metadata indexing.

Question 44 / 50

CORRECT ANSWER: B

Which statement about operating systems is correct?

- It replaces all application software
- **It acts as an interface between users, applications, and hardware**
- It is used only to browse websites
- It is stored only on external devices

Explanation: The OS abstracts hardware complexity and mediates access between applications and physical components.

Question 45 / 50

CORRECT ANSWER: A

Which of the following is NOT an example of system software?

- **Microsoft Excel**
- Android
- Windows
- Linux

Explanation: Excel is an end-user spreadsheet application; Android, Windows, and Linux are operating systems.

Question 46 / 50

CORRECT ANSWER: B

A USB flash drive is mainly used for:

- Displaying images on screen
- **Storing and transferring files**
- Processing calculations
- Connecting to internet

Explanation: USB drives provide portable, non-volatile solid-state storage.

Question 47 / 50

CORRECT ANSWER: B

Which combination correctly matches the device with its category?

- Mouse → Output, Speaker → Input
- **Keyboard → Input, SSD → Storage**
- Monitor → Input, HDD → Output
- Speaker → Storage, RAM → Input

Explanation: Keyboard provides input; SSD provides persistent secondary storage.

Question 48 / 50

CORRECT ANSWER: C

Why is system software generally necessary before application software can be used effectively?

- It prevents applications from accessing computer resources
- It replaces the need for hardware
- **It provides the basic environment and services applications depend on**
- It converts every application into hardware

Explanation: Applications rely on system software for memory management, process scheduling, I/O dispatch, and filesystem drivers.

Question 49 / 50

CORRECT ANSWER: C

A touchscreen on a smartphone acts as which type of device?

- Storage device
- Processing device
- **Input device**
- Power device

Explanation: The touch digitizer overlay registers capacitive touch coordinates and sends them into the processor as input events. (Note: visually it also serves output, but its touch-sensing component is an input mechanism).

Question 50 / 50

CORRECT ANSWER: C

Riya opens several applications simultaneously. The computer needs to allocate and schedule processor time to each. Which OS function is involved?

- Web browsing
- Memory allocation
- **Process management**
- File compression

Explanation: Process management and the CPU scheduler allocate CPU time slices and manage process states.

Contest 2: 25 Real Questions & In-Depth Solutions

Detailed analysis of all 25 questions covering Linux/Unix filesystem, chmod permissions, signals, process isolation, and CLI mechanics.

Question 1 / 25

NUMERIC ANSWER: 644

A results file is owned by the teacher. The teacher should be able to read and write the file, while the group and all other users should only be able to read it. Nobody should have execute permission. What three-digit numeric chmod permission value achieves exactly this?

- **644** (Owner: 4+2=6, Group: 4, Others: 4)

Explanation: Owner: Read (4) + Write (2) = 6. Group: Read (4) = 4. Others: Read (4) = 4. Combine digits: 644.

Question 2 / 25

CORRECT ANSWER: D

A browser tab crashes and shows an error, but other tabs continue working normally. What best explains this behavior?

- The browser automatically restarts the crashed tab in the background.
- The other tabs had already been saved to disk before the crash.
- The crashed tab had not actually been running as a process.
- **The browser isolates tab activity across processes, limiting the failure's impact.**

Explanation: Google Chrome runs each tab in an independent child process. A fatal crash in one child process terminates only that PID; siblings and parent continue unharmed.

Question 3 / 25

CORRECT ANSWER: D

In class demonstration, text typed in one terminal appeared in a second terminal after running `echo "hi" > /dev/pts/2`. What does this illustrate?

- Both terminals share one block of memory from the shell
- echo has a special mode for sending messages
- The two terminals were opened by the same parent shell
- **Terminals are represented as files, so writing works**

Explanation: Concrete manifestation of the Unix "Everything is a file" philosophy. Pseudo-terminals live under `/dev/pts/` and accept standard I/O redirection.

Question 4 / 25

CORRECT ANSWER: B

A newly created file shows `-rw-r--r--` in `ls -l`. Which permission does the owner not have?

- Read
- **Execute**
- Write
- The owner has all three

Explanation: The owner triplet is `rw-` (Read present, Write present, Execute absent).

Question 5 / 25

CORRECT ANSWER: A

A file `hello.txt` is made executable with `chmod +x hello.txt`, then renamed to `run_me.banana`. When running `./run_me.banana`, it executes successfully. Why?

- **On Unix-like systems, execute permission is stored in the file's permission metadata, not determined by its filename extension.**
- The shell uses the filename extension to determine executability.
- The .banana extension must be registered with the OS.
- Renaming a file automatically restores default permissions.

Explanation: Unix relies on metadata inode permission bits, unlike Windows where executability is tied to extensions like `.exe`.

Question 6 / 25

CORRECT ANSWER: A

The `$PATH` environment variable has been set to an empty string. Running `ls` fails with `command not found`. How can `ls` still be run?

- **By giving its full path: /bin/ls**
- By adding execute permission with chmod +x
- By running it through sudo instead
- By renaming the file to ls.exe first

Explanation: Emptying `$PATH` only disables automatic directory search. Giving the exact absolute path `/bin/ls` bypasses the search completely.

Question 7 / 25

NUMERIC ANSWER: 600

You keep private notes in a file that only you should be able to read and modify. No other users should have any permissions. What three-digit numeric chmod value achieves this?

- **600** (Owner: 4+2=6, Group: 0, Others: 0)

Explanation: Owner: Read (4) + Write (2) = 6. Group: None (0). Others: None (0).

Question 8 / 25

CORRECT ANSWER: B

You need to stop a running process, but you want to give it an opportunity to close its files and exit cleanly. Which command is appropriate?

- kill -9 — it terminates the process immediately
- **kill -15 — it requests that the process terminate**
- Either command — both signals behave identically
- Neither command

Explanation: `kill -15` issues SIGTERM, allowing the process to clean up buffers and gracefully exit. `kill -9` issues SIGKILL, which abruptly kills the process with zero cleanup.

Question 9 / 25

CORRECT ANSWER: D

Starting at `/` and running `cd ..` repeatedly leaves you at `/` every time. Why?

- Permission to move higher was denied
- cd .. is an alias that performs no action
- Directories above / exist but stay hidden
- **The root directory has no parent to move into**

Explanation: Root (`/`) is the topmost apex of the Unix directory tree; its parent pointer `..` points to itself.

Question 10 / 25

CORRECT ANSWER: B

Running `ls -l script.sh` shows `-rw-r--r--`. Running `./script.sh` returns `permission denied`. Which command makes it runnable while leaving existing read/write unchanged?

- sudo ./script.sh
- **chmod +x script.sh**
- chmod 777 script.sh
- chown root script.sh

Explanation: `chmod +x script.sh` selectively adds the execute bit without wiping or loosening existing read/write permissions.

Question 11 / 25

CORRECT ANSWER: B

After running `chmod 754 report.sh`, what permissions does the group hold?

- Read, write and execute
- **Read and execute**
- Read only
- Read and write

Explanation: Second digit is Group: 5 = 4 (Read) + 1 (Execute). Thus Group has Read and Execute (`r-x`).

Question 12 / 25

CORRECT ANSWER: C

Which directory is emptied automatically every time the machine reboots?

- /etc
- /bin
- **/tmp**
- /Applications

Explanation: `/tmp` stores ephemeral session data and is purged during system startup.

Question 13 / 25

CORRECT ANSWER: A

On a current macOS system, which shell is provided as the default command-line interface?

- **zsh**
- bash
- fish
- csh

Explanation: macOS switched its default shell from `bash` to `zsh` starting in macOS Catalina (2019).

Question 14 / 25

CORRECT ANSWER: C

Why cannot Chrome read data held in VS Code's memory?

- They are written in different programming languages.
- Their memory contents are encrypted separately by the OS.
- **The OS gives each process an isolated virtual address space.**
- They are scheduled on different CPU cores.

Explanation: Virtual memory architecture assigns each process a private address space; hardware MMU enforces strict isolation boundaries.

Question 15 / 25

CORRECT ANSWER: A

A classmate advises running `chmod 777` whenever a permission error appears. What is the best evaluation of this advice?

- **Poor — it grants read, write, and execute permissions to the owner, group, and others.**
- Sound — it is the quickest reliable fix available.
- Poor — 777 may only be applied by root.
- Sound — 777 makes a file read-only and safe.

Explanation: 777 dismantles all security restrictions and allows anyone to alter or execute the file. Always grant only the minimal permission required.

Question 16 / 25

CORRECT ANSWER: A

A directory holds 100 folders named `folder_1` through `folder_100`. Which single command removes them all?

- rm -r folder_*
- rm folder_
- rm -r folder_?
- rm -r folder_1

Explanation: The `*` wildcard matches any sequence of trailing characters (single, double, or triple digits), and `-r` recurses into directories.

Question 17 / 25

CORRECT ANSWER: A

Which command transfers ownership of a file to a different user?

- chown
- chmod
- mv
- cp

Explanation: `chown` (change owner) modifies file ownership; `chmod` modifies access permissions.

Question 18 / 25

CORRECT ANSWER: D

Which command reports the full filesystem path of the program that would run for a given command name?

- man
- pwd
- ps
- which

Explanation: `which <command>` queries `$PATH` and outputs the matching executable's absolute path (e.g. `which ls` → `/bin/ls`).

Question 19 / 25

CORRECT ANSWER: B

A production application repeatedly dies with an out-of-memory exception under load. What is the real remedy for the future?

- Add more CPU cores so work finishes sooner
- **Add more memory, or reduce what the app asks for**
- Change file permissions on the executable
- Terminate it with kill -9 whenever it slows

Explanation: CPU cores do not provide memory addresses. Out-Of-Memory can only be solved by allocating more physical RAM/swap or optimizing application memory footprint.

Question 20 / 25

CORRECT ANSWER: A

What does the `/dev` directory contain?

- **Files representing hardware devices**
- Developer tools, compilers and libraries
- Installed applications
- System configuration files

Explanation: `/dev` stands for "devices" and holds special device files (block and character devices) representing hardware peripherals and pseudo-devices.

Question 21 / 25

CORRECT ANSWER: D

Which command prints the absolute path of the directory you are currently working in?

- ls
- cd
- which
- **pwd**

Explanation: `pwd` stands for Print Working Directory.

Question 22 / 25

CORRECT ANSWER: B

Permissions are given numeric values: read is 4, write is 2, execute is 1. Which single digit grants all three at once?

- 3
- **7**
- 6
- 9

Explanation: 4 (Read) + 2 (Write) + 1 (Execute) = 7.

Question 23 / 25

CORRECT ANSWER: B

Assuming Google Chrome browser is a parent process, one tab of Chrome corresponds to what, in operating-system terms?

- One parent process
- **One child process**
- One thread of main process
- One separately installed program

Explanation: Chrome's multi-process model allocates one child process per tab for sandbox isolation and fault containment.

Question 24 / 25

CORRECT ANSWER: B

Why does a newly created file deliberately lack execute permission?

- It keeps the file smaller on disk
- **It stops unvetted content being run by accident**
- Only the root user is allowed to create executables
- Execute permission applies only to directories

Explanation: Security principle of least privilege: arbitrary created or downloaded files cannot run without explicit affirmative `chmod +x` action by the user.

Question 25 / 25 CORRECT ANSWER: C

Google Chrome can be installed once on disk. When launched, multiple processes are created. What is the fundamental difference between a program and a process?

- Program is source code; process is machine language.
- Program needs CPU; process needs only memory.
- **A program is a stored set of instructions; a process is an instance of that program that is currently executing, along with its execution state and allocated resources.**
- Program belongs to user; process belongs to OS.

Explanation: Program = passive stored code file on disk. Process = active running instance loaded into RAM with memory space, threads, and execution context.